

Rama Gangumalla

Data Engineer

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Summary

Data Engineer with 4+ years of experience developing ETL/ELT pipelines, designing data models, and delivering analytics using Python, PySpark, SQL, and Power BI. Recent work includes production pipelines on Azure Data Lake Storage and Databricks that handle millions of test results per day, and an internal Angular and FastAPI application that connects Databricks workflows to test planning, Excel intake, and interactive performance reporting.

Skills

Languages: Python, SQL, TypeScript, C/C++

Cloud & Storage: Azure (Data Factory, Data Lake Storage Gen2, Event Hubs, Synapse Analytics)

Processing & Lakehouse: Databricks, Apache Spark, PySpark, Medallion Architecture

Databases: PostgreSQL, MySQL, Redis

Data Engineering: ETL/ELT, OLAP, Data Modeling, Data Warehousing, Performance Tuning

Application Stack: Angular, FastAPI, SQLAlchemy, REST APIs

BI & Reporting: Power BI (Dataflows, Datamarts), Azure Analysis Services, Databricks SQL, OBIEE

Tools & Practices: Linux, Git, Docker, Docker Compose, GitHub Actions, Okta, Agile, Metadata Management

Professional Experience

Data Engineer, AMD

Apr 2025 - Present

- Designed data models and built ETL/ELT pipelines in Databricks that ingested millions of performance-test results per day from internal systems into Azure Data Lake Storage and transformed them into curated, analytics-ready layers.
- Processed large volumes of test data with PySpark on Databricks using notebooks and scheduled jobs, and optimized those transformations to reduce data-processing time by 25% versus the prior Spark jobs.
- Built a three-tier internal web application (Angular, FastAPI, PostgreSQL, Redis) for discovering Databricks workflows, planning tests, generating Excel intake sheets, and submitting them to the lab scheduling system through a single interface.
- Designed the FastAPI backend with cookie-based JWT authentication, CSRF protection, Okta SSO, API keys, and tenant-scoped data isolation enforced in the service layer.
- Containerized the stack with Docker Compose and set up GitHub Actions CI for backend tests and frontend builds.
- Implemented an admin-triggered Databricks sync that loads workflow, task, station, and team dimension tables in parallel into a local cache, with Redis-backed progress tracking and REST status endpoints.
- Delivered general matrix multiply (GEMM) benchmark reporting using a Databricks SQL view, querying current data on each request and exposing date and column filters, KPI cards, and a configurable chart builder.
- Developed Power BI dashboards on curated test metrics and presented insights to performance-engineering teams.

Data Engineer, Wayfair

Oct 2023 - Mar 2025

- Built end-to-end customer clickstream pipelines ingesting web and mobile SDK events through Azure Event Hubs to support near real-time funnel analytics and downstream reporting.
- Managed Bronze, Silver, and Gold data layers in Azure Data Lake Storage Gen2, supporting daily incremental data for analytics and machine-learning workloads.
- Developed PySpark and SQL transformations to calculate funnel conversion, drop-off, and session-abandonment metrics across checkout journeys.
- Implemented data cleansing and validation checks across ingestion and transformation stages to maintain data quality and consistency in downstream reporting.
- Loaded analytics-ready and ML-ready datasets into Azure Synapse Analytics and optimized reporting models through partitioning, indexing, and join optimization, reducing reporting-query runtime by 30% versus the prior unoptimized models.
- Gathered reporting requirements from business stakeholders and built Power BI dataflows and datamarts over multiple source systems to establish a single source of truth for reporting across business teams.
- Developed large-scale analytical models in Power BI datasets and Azure Analysis Services, working alongside engineers and analysts to provide a semantic layer for visualization, report development, and ad hoc analysis.

Graduate Research and Teaching Assistant, Kennesaw State University**Aug 2022 - May 2023**

- Assembled and configured V2X devices for traffic-signal monitoring and emergency-vehicle preemption, and modeled related traffic scenarios using Simulation of Urban Mobility (SUMO).
- Developed a C++ application to detect potholes from image data, store detection records, and generate SQL-based reports and visualizations of identified potholes.
- Created Power BI dashboards to monitor emergency-vehicle flow by area and traffic-signal status.
- Developed a SUMO vehicle-flow simulation to evaluate traffic impacts for a highway-exit construction project.

System Engineer, Tata Consultancy Services**Feb 2020 - Dec 2021**

- Operated and maintained ETL jobs and OLAP reporting for health-insurance metrics, ensuring scalability and operational stability.
- Partnered with insurance users, claims analysts, and senior engineers to map end-to-end data flow across policy, claims, billing, and customer systems, and translated reporting needs into technical data pipelines.
- Developed interactive Power BI dashboards and OBIEE reports for multidimensional datasets across insurance and HR domains.
- Supported migration of on-premises insurance data to Azure using Azure Data Factory and Azure Data Lake Storage Gen2, enabling cloud-based processing of large policy and claims datasets for analytics.

Education

Master of Science in Computer Science, Kennesaw State University, Marietta, GA

Jan 2022 - May 2023

Bachelor of Technology in Mechanical Engineering, R.V.R. & J.C. College of Engineering, Guntur, India

Jun 2015 - May 2019

Certifications

AWS Certified Data Engineer - Associate

Microsoft Certified: Azure Database Administrator Associate